

Differential effect of prior paradoxical sleep deprivation on conditioned taste aversion, neophobia and attenuation of neophobia to solid food in rats

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The effect of paradoxical sleep deprivation (PSD) on subsequent acquisition of conditioned taste aversion (CTA) to liquid diets is confounded by the uncertain level of thirst when using the water tank procedure. This difficulty is eliminated when examining CTA and attenuation of neophobia (AN) to solid diets. Adult male rats ($n = 100$) were habituated to receive their daily ration of food during a 30-min stay in a box equipped with a row of 10 feeders baited with 2–3 g pieces of moist standard diet. 24-h PSD increased neophobic rejection of novel sweet food (with added 5% saccharose), but did not influence intensity of CTA elicited by LiCl poisoning. Addition of a bitter tasting red food dye to the sweet food caused marked neophobia which was enhanced by preceding PSD. Association of this unpalatable food with LiCl elicited strong CTA which extinguished faster in the PSD-pretreated animals. On the other hand, preacquisition PSD did not influence AN to the same stimulus. Sweet food with added blue dye elicited only mild neophobia which was enhanced by preceding 24-h PSD. Preacquisition PSD did not influence AN but significantly increased CTA to blue sweet food. It is concluded that PSD can either enhance or weaken CTA and that this complex effect on food selection learning cannot be explained by PSD-induced reduction of fear.

INTRODUCTION

The disruptive effect of paradoxical sleep deprivation (PSD) on subsequent learning remains a controversial issue¹⁵. Clear impairment was only observed in 6 out of 15 studies reviewed by Smith¹⁶. The effect obviously depends of the nature of the task, method of PSD induction, PSD duration as well as on a number of procedural and organismic factors. Short lasting PSD (24 h or less) elicited the most remarkable learning deficit when applied immediately before acquisition of conditioned taste aversion (CTA). Danguir and

Nicolaidis⁵ exposed thirsty rats to water-tank PSD before offering them a novel fluid (isotonic LiCl solution) which served simultaneously as the conditioned (CS) and unconditioned stimulus (US). Even a 7.5-h PSD significantly decreased subsequent aversion to the salty taste. Venkatakrishna Bhatt and Bureš¹⁹ confirmed this finding using a modified PSD technique. In order to prevent thirsty rats from drinking the surrounding water, the pedestals were placed onto an electrified grid floor and foot shocks forced the animals to return to the elevated platforms. After 24-h PSD, saccharin (CS) drinking followed by LiCl

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poisoning did not elicit CTA. The results of both studies can be influenced, however, by such unknown factors as the uncertain level of water deprivation in the first case and the variable stress caused by electrocutaneous stimulation in the second case. The purpose of the present study was to examine the PSD effect on CTA acquisition under conditions better compatible with the standard water tank technique. This was made possible by using hungry rather than thirsty rats and by offering the taste stimulus not in liquid form but as solid food¹. A secondary goal of the present experiments was to test the assumption that the impairment of CTA acquisition is due to PSD-induced reduction of fear by examining the PSD effect on neophobia and on the formation of the gustatory trace underlying attenuation of neophobia (AN), i.e. increased intake of diets, the ingestion of which was not followed by adverse visceral consequences⁷.

GENERAL METHODS

Subjects

Male hooded rats of the Long-Evans strain were obtained from the breeding colony of the Institute. The 3-month-old animals were housed in group cages (5 animals per cage) in a room with constant temperature and natural lighting. Water was freely accessible in the home cage but food was only available during the 0.5-h daily experiment. Each animal received 20 g of standard food pellets on the weekends.

Apparatus

The plastic feeding box (26 × 15 × 15 cm) had a wire mesh ceiling. A feeder placed along one of the shorter walls of the box was a flat perspex block (12 × 6 × 2 cm) with 10 hemispheric recesses (1.7 cm in diameter) in the upper surface. Each hole could accommodate a 2–3 g piece of soft rat diet.

PSD was elicited by placing the rats into plastic buckets (21 cm in diameter and 27 cm high) which could be closed with wire mesh lids. Cylindrical pedestals (7.5 cm high and 7.0 cm in diameter) were fixed in the center of the bottom. The buckets were filled with 20 °C water to a height of

6.5 cm, i.e. 1 cm below the rim of the pedestals. EEG signs of PSD elicited on pedestals of this type in the strain of rats employed were demonstrated in an earlier study¹⁸.

Gustatory stimuli

Powdered rat diet (100 g) was mixed with water (50 ml) and with various additives (5 g of saccharose and/or 1 g of food dye). The soft paste was divided into ball-shaped pieces (2–3 g) fitting into the holes of the feeder. The weight of the food was established before and after the experiment with 0.1 g accuracy.

Procedure

Rats were first adapted during 5–6 days to the feeding conditions. They were placed into the individual feeding boxes with the feeder containing 10 pieces of standard diet and allowed 30 min of eating. After food consumption had stabilized, the rats were divided into groups equated for food intake. Some groups were subjected to 24-h PSD and immediately afterwards offered novel food (sweet and/or coloured food). In CTA or AN experiments, ingestion of the novel food was followed by i.p. administration of LiCl or NaCl, respectively (0.15 mol/l, 2% body weight). On the next day, the animals were offered standard diet. Retention was tested in a single-choice or a two-choice retrieval test with the feeder containing 10 pieces of sweet food in the first case or 5 pieces of standard food and 5 pieces of sweet coloured food in the second case. Consumption of each type of food was established as the difference between the weight of food placed into the feeder and what remained there after termination of the retrieval test. Modifications of this basic procedure are described in connection with the individual experiments.

EXPERIMENT 1

The PSD-induced impairment of CTA acquisition was established with the single bottle technique^{5,18}. We attempted, therefore, to examine the PSD effect on the acquisition of CTA to solid food under conditions of single-choice retrieval.

Method

The procedure described in General Methods was followed. Two groups of 10 rats each were adapted to the feeding schedule during 6 days. PSD was started immediately after the sixth feeding session and terminated before the feeding session on Day 7 in the group PSD/CTA. Both groups received sweet food on Day 7 and an LiCl injection immediately after exposure to the novel food. Both groups received standard food on Day 8 and the sweet food on Day 9.

Results and discussion

Fig. 1 shows that the two groups had similar food intake on Day 6. Two-tailed *t*-tests for paired values indicated that the consumption of sweet food was significantly increased on Day 7 in the CTA group ($t_9 = 2.8$, $P < 0.05$) but not in the PSD/CTA group ($t_9 = 0.9$, n.s.). Comparison of the Day 7 data showed decreased consumption of the novel food after PSD ($t_{18} = 2.31$, $P < 0.05$).

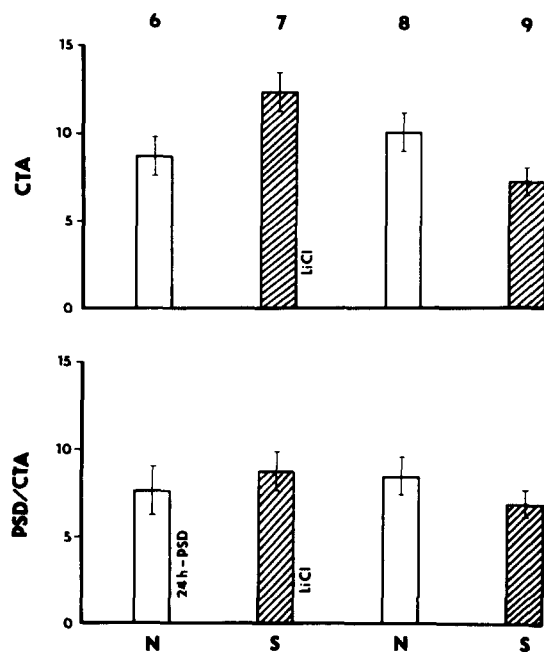


Fig. 1. The effect of 24-h PSD on the acquisition of CTA to sweet food. Abscissa: days of experiment. Ordinate: average ($\pm$ S.E.M.) food intake in grams. N, standard diet (empty columns); S, standard diet with added 5% saccharose (shaded columns); 24-h PSD, PSD starting after session 6 and terminating before session 7; LiCl, intraperitoneal injection of 0.15 mol/l LiCl, 2% body weight.

Retrieval test on Day 9 showed significant decrease of sweet food intake (as compared with Day 7) in both groups. The drop was more expressed in the CTA group ($t_9 = 5.2$, $P < 0.01$) than in the PSD/CTA group ($t_9 = 2.1$, $P < 0.05$). Comparison of the paired differences between sweet food consumption during acquisition and retrieval indicated that the sweet food intake decreased in the CTA group significantly more than in the PSD/CTA group ($t_{18} = 2.45$, $P < 0.01$). Absolute food intake during the retrieval test was similar in both groups.

The results of Expt. 1 suggest that the palatability of sweet food counteracted neophobia and increased food intake in the CTA group. Lower consumption of sweet food in the PSD/CTA group may indicate PSD-induced enhancement of neophobia. The retrieval test revealed only weak CTA, the evaluation of which was confounded by the PSD effect on neophobia. It seems that the gustatory stimulus was not sufficiently salient to support formation of CTA which could be detected under the conditions of single-choice retrieval.

EXPERIMENT 2

Two-bottle tests are more sensitive⁹ than single-bottle tests when establishing weak taste preferences. In order to improve CTA detection, retrieval was tested in Expt. 2 by offering the animals a choice between standard food and sweet coloured food previously associated with LiCl poisoning. Addition of the food dye not only allowed reliable identification of the standard and novel foods but also served as an additional gustatory stimulus increasing the salience of the novel taste. Two additional groups were included to assess the effect of PSD on AN.

Method

The procedure described in General Methods was followed. After 5 days of habituation to the feeding conditions, rats were assigned to 4 groups of 10 animals equated for food intake: CTA, PSD/CTA, AN and PSD/AN. The PSD groups spent the next 24 h on the pedestals in the water-filled buckets. On Day 6 all groups were offered

red sweet food (5% saccharose and 1% of red food dye). Immediately after 30-min feeding the CTA groups were injected with LiCl and the AN groups with NaCl. After receiving standard food on Day 7, all animals were given a choice between standard food and red sweet food on Day 8. Standard food and sweet food (without red dye) were given to all animals on Days 9 and 10, respectively.

ly. The second retrieval test was performed on Day 11.

Results and discussion

The critical stages of Expt. 2 are illustrated by Fig. 2. The 4 groups did not differ according to standard food consumption on Day 5. The intake of red sweet food decreased in all 4 groups on

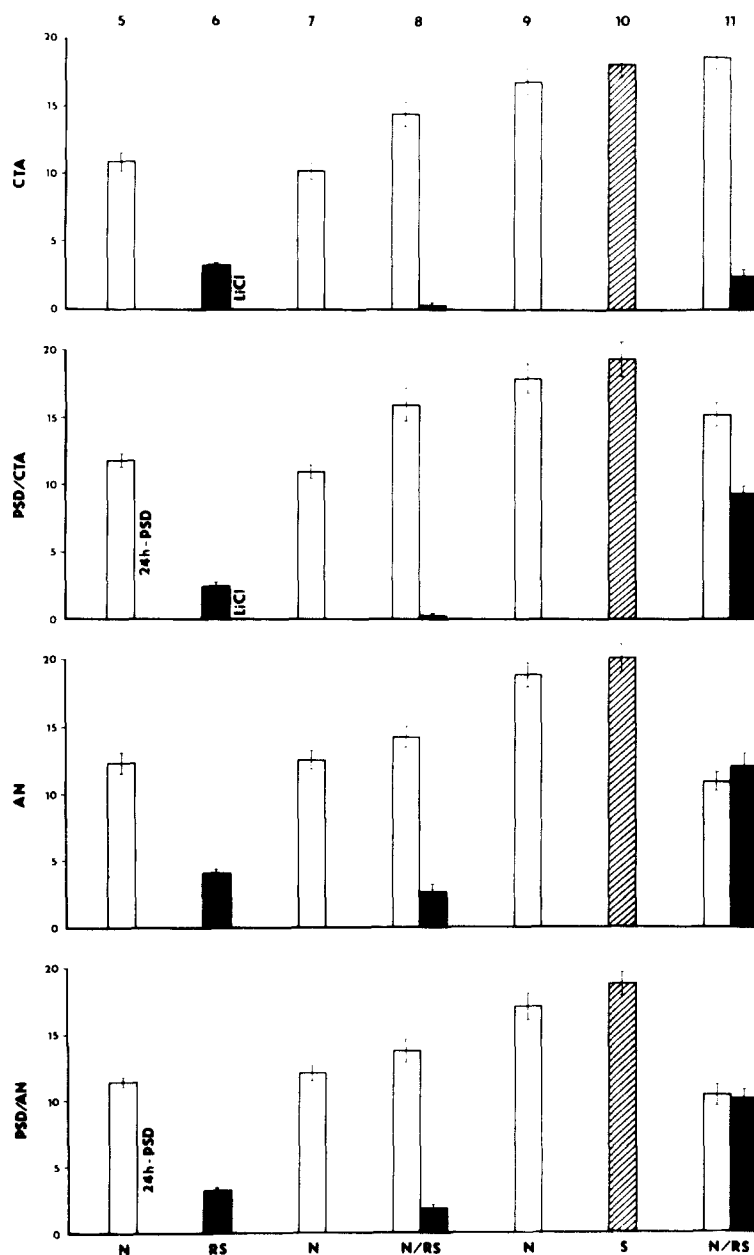


Fig. 2. The effect of 24-h PSD on the acquisition of CTA or AN to red sweet food. RS, sweet food with added 1% of red food dye (black columns). Other description as in Fig. 1.

Day 6 as compared with Day 5 ($P < 0.01$, two-tailed t -test, paired comparisons). One-way analysis of variance showed significant between-group differences ($F_{3,36} = 7.4$, $P < 0.01$) due to decreased consumption of novel food in the PSD groups. There was no between-group difference in the standard food intake on Day 7. The two-choice retrieval test on Day 8 showed almost complete avoidance of red sweet food in both CTA groups and substantial aversion in the AN groups. One-way analysis of variance of the average ratios of red sweet food to the total food consumed indicated a significant treatment effect ($F_{3,36} = 31.8$, $P < 0.001$). Newman-Keuls multiple comparisons showed that the CTA groups consumed significantly less ($P < 0.01$) red sweet food than the AN groups. No other differences were significant. Consumption of standard food and of sweet standard food was increased on Days 9 and 10 similarly in all groups. A second two-choice retrieval test on Day 11 revealed low intake of red sweet food in the CTA groups and greatly increased consumption of this food in the AN groups. One-way analysis of variance performed on the red sweet food/total food consumption ratios revealed significant treatment effect ($F_{3,36} = 4.2$, $P < 0.025$). Newman-Keuls multiple comparisons showed that the CTA group was significantly different from the PSD/CTA group ($P < 0.01$) and from both AN groups. No other difference reached significance.

The above results indicate that addition of the red food dye had considerably increased neophobic rejection of the novel food and that the effect was further enhanced by preceding PSD. Association of the red sweet food with LiCl poisoning elicited strong CTA which could not be further enhanced by preacquisition PSD because of a floor effect. On the other hand, PSD did not significantly influence AN for the red sweet food which was similarly rejected by both AN and PSD/AN groups. Since sweet food without dye was eagerly consumed by all groups on Day 10, the slightly bitter taste of the red food dye containing 'amarant' was obviously the salient component of the complex gustatory stimulus. Experience with the sweet taste probably enhanced extinction of CTA to the complex stimulus mani-

fested by weaker CTA on Day 11. A significant difference between the CTA and PSD/CTA groups points toward the conclusion that PSD preceding the first presentation of the red sweet food resulted in the formation of a weaker CTA which was less resistant to extinction.

EXPERIMENT 3

Comparison of Expts. 1 and 2 suggested differential effect of PSD on formation of CTA to taste stimuli eliciting strong and weak neophobia, respectively. This conclusion was verified in the next series of experiments using a food dye of neutral taste.

Method

The same procedure as in Expt. 2 was followed but a blue food dye (1%) was used instead of the red one because among several food dyes tested it elicited minimum neophobia. The experiment was terminated after the first retrieval test on Day 8.

Results and discussion

As shown in Fig. 3, there was no difference between the 4 groups on the last day of habituation. The intake of the novel blue sweet food was significantly increased in the CTA and AN groups ($t_9 = 1.9$, $P < 0.05$ and $t_9 = 2.5$, $P < 0.025$ respectively), but not in the PSD/CTA and PSD/AN groups. One-way analysis of variance of the blue sweet food consumption data on day 6 did not yield significant between-group differences ($F_{3,36} = 2.06$, n.s.). Standard food consumption was similar in all groups on Day 7. Retrieval test on Day 8 showed clear preference for the blue sweet food in both AN groups, clear avoidance of this food in the PSD/CTA group and an intermediate result in the CTA animals. One-way analysis of variance of the blue sweet food/total food intake ratios showed significant between-group differences ($F_{3,36} = 15.8$, $P < 0.01$). Newman-Keuls multiple comparisons confirmed significant difference between the PSD/CTA group and all remaining groups ($P < 0.01$) as well as between the CTA group and

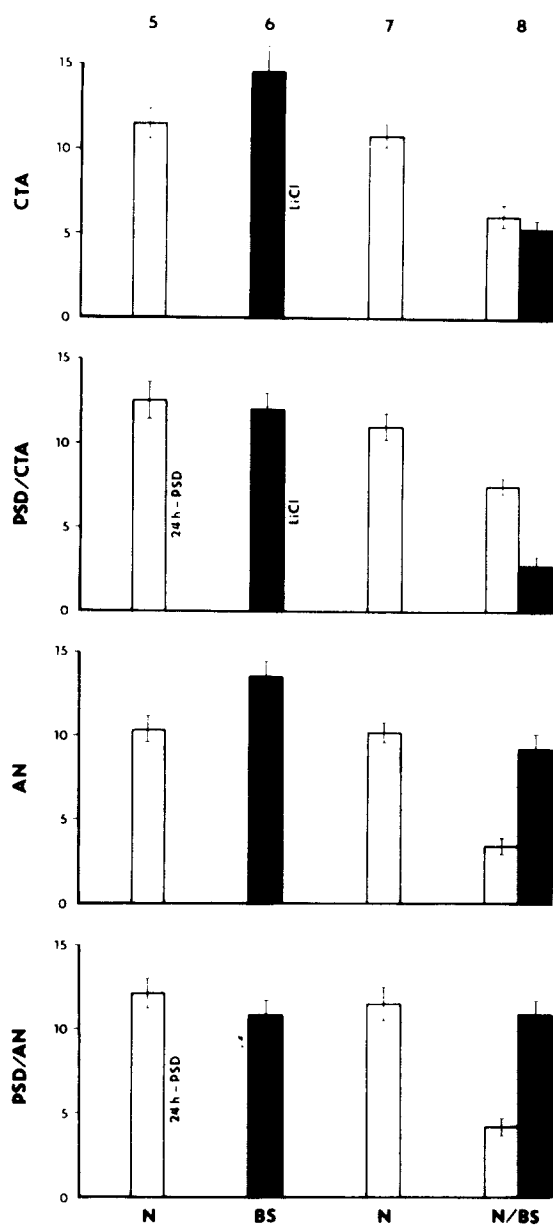


Fig. 3. The effect of 24-h PSD on the acquisition of CTA and AN to blue sweet food. BS, sweet food with added 1% of blue food dye. Other description as in Fig. 1.

both AN groups ($P < 0.01$). There was no difference between the AN and PSD/AN groups.

The results of Expt. 3 resemble those of Expt. 1, but the use of the two-choice retrieval test made it possible to assess the PSD effect on CTA acquisition more accurately. PSD preceding presentation of the novel gustatory stimulus increased neophobia and enhanced CTA formation but did not affect the development of AN.

GENERAL DISCUSSION

The results of the present study indicate that processing of the taste properties of solid food and of liquid diets is differentially affected by PSD. Strong CTA was obtained to food eliciting marked neophobia (Expt. 2). Reduction of CTA strength by preacquisition PSD was manifested in this case by faster CTA extinction. This result is consonant with the earlier reports^{5,19}, although the PSD-induced impairment of CTA acquisition was less pronounced in the present experiments. On the other hand, CTA to palatable food eliciting no initial neophobia (Expt. 3) was relatively weak and its strength was increased by preacquisition PSD. Similar results were obtained in Expt. 1, but the single-choice retrieval test was not sensitive enough to demonstrate the weak CTA with sufficient clarity.

Weak CTA to palatable food is probably caused by latent inhibition¹⁷. Standard rat diet contained 60% ground wheat and 9% wheat bran and its taste was not much altered by the addition of 5% sucrose. It is well known that it is difficult to form CTA to familiar taste stimuli, the presentation of which has never before been followed by sickness⁸.

On the other hand, new flavours are more conspicuous when presented in liquid form to rats used to receive their daily ration of fluid as tap water.

The weakening of CTA to unpalatable gustatory stimuli presented after PSD can reflect changes of emotionality. It is conceivable that formation of the short-term gustatory trace is enhanced by fear eliciting properties of the particular flavour. PSD-induced reduction of fear in rats is indicated by increased exploration and decreased defecation, urination and freezing in the open field^{10,14,15}. PSD increases novelty preference in a Y-maze¹³ and diminishes neophobia towards novel objects in the open field¹². This is not the case with the neophobia to novel food which was not diminished but rather enhanced by PSD in the present study. It seems that removal from the stressful water tank situation makes the rats less sensitive to other environmental risks but does not reduce visceral fear. The finding that PSD did not

influence AN to the taste stimuli examined, but differentially affected formation of CTA to the same stimuli, is consonant with the view^{2,11} that neophobia and CTA are independent phenomena. Such a conclusion is also supported by the evidence indicating that AN is more sensitive than CTA to functional disruption^{3,4} and to lesions⁶.

The above considerations lead to the conclusion that the 24-h PSD does not prevent recognition of the novel taste stimuli but rather increases their neophobic rejection. The gustatory trace formed during presentation of the novel taste is not significantly influenced by preceding PSD because it supports similar AN in both PSD-treated and control animals. PSD enhancement of weak CTA corresponds to increased neophobia and is consonant with the well-established view that flavours eliciting marked neophobia produce strong CTA⁷.

On the other hand, PSD-induced weakening of strong CTA cannot be attributed to decreased salience of the gustatory stimulus, but is probably due to decreased associability of the gustatory trace with subsequent poisoning. Further research is required to assess the relative contributions of PSD and of other concomitants of the pedestal procedure to the above effects and to specify the conditions conducive to CTA enhancement or CTA disruption.

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